

The Quantum Infrastructure of the Standard Nuclear oliGARCHy

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Abstract

This paper develops a theory of quantum infrastructure within the Standard Nuclear oliGARCHy (SNoG). It extends a general theory of infrastructure and connects quantum computing, human capital, cryptographic security, finance, economic development and strategic capability. The central proposition is that quantum infrastructure should not be understood merely as a collection of quantum computers. It is an integrated system consisting of physical quantum hardware, error correction, algorithms, software, scientific human capital, energy, communications, financial resources and institutional capacity. The paper uses real-options theory, network economics, welfare economics, international political economy, strategic studies, statistics, econometrics and artificial intelligence to construct a compact analytical framework. The resulting system is interpreted as a strategic infrastructure option whose value depends upon technological progress, human-capital formation, resilience and the timing of adoption.

The paper ends with “The End”

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1 Introduction

The Standard Nuclear oliGARCHy requires an infrastructure theory capable of representing not only roads, ports, energy systems and communications but also emerging technological infrastructures.

The general infrastructure framework treats infrastructure as a portfolio of real options, complementary assets and networks. The present paper identifies quantum infrastructure as a distinct layer because quantum capability introduces a technological state variable with potentially discontinuous implications for cryptography, computation, finance and strategic autonomy.

The motivating technological literature includes recent work on fault-tolerant quantum computation with reconfigurable neutral-atom architectures and high-rate quantum error-correcting codes [4, 6, 7].

The economic motivation follows the connection between human-capital policy and technological capacity developed in the Italy-oriented analysis [5, 1].

2 Definition of Quantum Infrastructure

We define quantum infrastructure as the integrated system

$$\mathcal{Q}_t = (Q_t, H_t, E_t, C_t, S_t, F_t, I_t), \quad (1)$$

where Q_t is physical quantum-computing capacity, H_t is quantum-skilled human capital, E_t is energy and environmental capacity, C_t is classical communication and computing infrastructure, S_t is software and algorithmic capacity, F_t is financial capacity and I_t is institutional capacity.

The important distinction from ordinary infrastructure is therefore functional rather than merely physical.

$$\text{Quantum Infrastructure} \neq \text{Quantum Hardware Alone.} \quad (2)$$

A state possessing advanced hardware but lacking engineers, error-correction expertise, algorithms, electricity, communications or institutional capacity does not possess equivalent effective quantum capability.

3 The Quantum Capability Function

Let effective quantum capability be

$$\mathcal{K}_t^Q = K_Q(Q_t, H_t, E_t, C_t, S_t, F_t, I_t). \quad (3)$$

A simple multiplicative representation is

$$\mathcal{K}_t^Q = Q_t^\alpha H_t^\beta E_t^\gamma C_t^\delta S_t^\epsilon F_t^\zeta I_t^\eta, \quad (4)$$

with nonnegative parameters.

The multiplicative form emphasizes complementarity. If any critical component approaches zero, effective capability can fall sharply even when other components are highly developed.

4 Quantum Infrastructure as a Real Option

Quantum investment is characterized by uncertainty, technological progress, large fixed costs and potentially irreversible expenditure. It is therefore naturally represented as a real option [8, 9].

Let the value of quantum capability be

$$P_t^Q = P^Q(\mathcal{K}_t^Q, R_t, X_t), \quad (5)$$

where R_t represents strategic and technological risk and X_t represents the wider economic state.

A state may delay investment until technological uncertainty falls, invest immediately, expand existing infrastructure or abandon a programme.

The corresponding option value can be written abstractly as

$$V_t^Q = \max_{\tau} \mathbb{E}_t \left[e^{-r(\tau-t)} (P_{\tau}^Q - K_{\tau}^Q) \right], \quad (6)$$

where τ is the exercise time.

5 Human Capital as Quantum Infrastructure

Quantum capability requires specialized human capital.

Let s_t denote human-capital policy and H_t the effective quantum-STEM human-capital stock. A stylized accumulation equation is

$$\dot{H}_t = \alpha s_t - \delta H_t. \quad (7)$$

Effective quantum capability then evolves according to

$$\dot{Q}_t = \beta H_t - \gamma Q_t. \quad (8)$$

This creates a two-stage mechanism:

$$s_t \longrightarrow H_t \longrightarrow Q_t \longrightarrow \mathcal{K}_t^Q. \quad (9)$$

The Italy-specific analysis motivating this paper explicitly models child-care subsidy policy as an input into human capital and subsequently into quantum capability [1, 5].

6 Quantum Infrastructure and Cryptography

The principal strategic distinction between classical and quantum infrastructure is that sufficiently capable quantum computers threaten important public-key cryptographic systems.

Shor's algorithm provides polynomial-time quantum algorithms for integer factorization and discrete logarithms [10].

For an idealized computational-security representation,

$$T_{\text{classical}} \gg T_{\text{quantum}} \quad (10)$$

for sufficiently large cryptographic instances under the relevant quantum algorithmic assumptions.

Grover's algorithm instead provides a quadratic speedup for unstructured search [11]:

$$T_{\text{Grover}} = O\left(2^{n/2}\right). \quad (11)$$

The distinction is therefore important:

$$\text{Shor} \rightarrow \text{Public-Key Cryptography}, \quad (12)$$

while

$$\text{Grover} \rightarrow \text{Generic Search}. \quad (13)$$

The present paper concerns defensive infrastructure planning and post-quantum migration rather than operational exploitation of cryptographic systems.

7 Error Correction as Infrastructure

Fault tolerance converts physical qubits into logical computational resources.

Let p be the physical error rate and p_L the logical error rate. A stylized error-correction relation is

$$p_L = f(p, d, R), \quad (14)$$

where d represents code distance and R represents the relevant code architecture.

The technological literature motivating the present paper investigates high-rate lifted-product quantum low-density parity-check codes and reconfigurable neutral-atom architectures [4].

The economic implication is straightforward: improvements in error correction reduce the physical-resource cost of a desired logical capability.

$$\frac{\partial Q_{\text{physical}}}{\partial p_L} > 0. \quad (15)$$

Consequently, advances in mathematics and quantum error correction can produce infrastructure-equivalent improvements without constructing proportionally more physical machines.

8 Quantum Infrastructure as a Network

Quantum infrastructure is also a network.

Let

$$G_t^Q = (V_t^Q, E_t^Q, W_t^Q) \quad (16)$$

represent quantum laboratories, data centres, universities, semiconductor facilities, energy systems, communications infrastructure and financial institutions.

The value of the network can be represented as

$$C_t^Q = \sum_{i,j} A_{ij,t}^Q I_{i,t}^Q I_{j,t}^Q. \quad (17)$$

The graph representation implies that quantum capability is partly determined by connectivity.

A university, quantum laboratory and industrial facility may therefore produce more value jointly than separately.

9 Complementarity

Suppose two quantum infrastructure components have joint value

$$V(Q_1, Q_2) = V_1(Q_1) + V_2(Q_2) + \phi Q_1 Q_2. \quad (18)$$

Then

$$\frac{\partial^2 V}{\partial Q_1 \partial Q_2} = \phi. \quad (19)$$

Hence

$$\phi > 0 \implies \text{quantum infrastructure complementarity.} \quad (20)$$

This provides a formal reason why hardware, algorithms, human capital, energy and institutions should be treated as a system.

10 Finance and Investment

Let the cost of quantum infrastructure be K_t^Q and its expected future economic value be B_t^Q . A simplified net present value is

$$NPV_t^Q = \mathbb{E}_t \left[\sum_{s=t+1}^T \frac{B_s^Q - C_s^Q}{(1+r)^{s-t}} \right] - K_t^Q. \quad (21)$$

However, the real-options formulation is superior whenever investment can be delayed.

Let the investment threshold be Q^* . Exercise occurs when

$$\mathcal{K}_t^Q \geq Q^*. \quad (22)$$

The threshold itself is endogenous to technological uncertainty, financing conditions and strategic competition.

11 Econometrics

The empirical relationship between quantum infrastructure and economic outcomes may be estimated through

$$Y_{c,t} = \alpha_c + \delta_t + \beta \mathcal{K}_{c,t}^Q + \gamma' X_{c,t} + u_{c,t}, \quad (23)$$

where c indexes countries or SNOG districts.

Possible dependent variables include productivity, research output, high-skilled employment, exports, financial development, cybersecurity investment and innovation.

Endogeneity is likely because technologically advanced economies may simultaneously possess greater quantum capability and higher economic performance.

Accordingly, empirical work may require instrumental variables, difference-in-differences, synthetic controls, panel estimators or structural models.

12 Statistics and Bayesian Updating

Let D_t represent information arriving by period t . The posterior distribution of a technological hypothesis H is

$$\Pr(H \mid D_t) = \frac{\Pr(D_t \mid H) \Pr(H)}{\Pr(D_t)}. \quad (24)$$

This is particularly important for quantum technology because forecasts of hardware requirements can change rapidly as algorithms, error-correcting codes and physical architectures improve.

A Bayesian infrastructure planner should therefore update the exercise probability rather than treating an old technology forecast as permanent.

13 Artificial Intelligence and Machine Learning

AI/ML can enter the quantum-infrastructure architecture through at least four channels:

1. quantum-error-correction optimization;
2. algorithm and circuit optimization;
3. scientific discovery and materials research;
4. forecasting of technological and economic variables.

A general predictive model is

$$\hat{Y}_t = f_\theta(X_t). \quad (25)$$

For sequential technological data, recurrent, state-space and transformer architectures can be considered [12, 13].

For infrastructure networks, graph neural networks are natural candidates:

$$h_v^{(l+1)} = \phi \left(h_v^{(l)}, \sum_{u \in \mathcal{N}(v)} A_{uv} h_u^{(l)} \right). \quad (26)$$

The model should remain subordinate to empirical validation. Prediction accuracy does not by itself establish causality.

14 Human Capital, Family Policy and Welfare

The human-capital mechanism has a welfare dimension.

Let welfare be

$$W_t = U(C_t, L_t, H_t, Q_t) - C_t, \quad (27)$$

where consumption, labour supply, human capital and technological capacity enter social utility.

A human-capital policy may therefore have several effects:

$$s_t \rightarrow L_t \rightarrow H_t \rightarrow Q_t \rightarrow Y_t. \quad (28)$$

The Italy-specific analysis uses child-care policy as an example of such a channel, drawing on the kinked subsidy framework of Pavoni and Ho [5, 1].

The relevant welfare conclusion is conditional rather than universal: policy should be evaluated according to its costs, distributional effects, labour-supply response and resulting human-capital gains.

15 Physics, Chemistry and Materials Science

Quantum infrastructure depends fundamentally upon physical systems.

The relevant physical layers include quantum states, control systems, measurement, cooling or trapping systems, optical systems and error processes. Neutral-atom architectures are one example of a physical implementation [7, 4].

Chemistry and materials science enter through the design of physical components, optical systems, materials, fabrication and environmental stability.

The economic representation is therefore

$$\text{Materials} \rightarrow \text{Physical Qubits} \rightarrow \text{Logical Qubits} \rightarrow \text{Algorithms} \rightarrow \text{Economic Capability}. \quad (29)$$

The chain demonstrates why the quantum economy is inherently interdisciplinary.

16 Biology and Human Systems

Biology enters principally through human capital rather than through a claim that quantum infrastructure is itself biological.

Education, fertility, childcare, health and labour-force participation affect the supply of scientists and engineers.

A stylized human-capital stock is

$$H_t = H_t^{\text{education}} + H_t^{\text{experience}} + H_t^{\text{migration}} - H_t^{\text{attrition}}. \quad (30)$$

Thus demographic and biological processes can affect the long-run supply of quantum expertise.

17 Psychology and Collective Decision-Making

Quantum infrastructure requires expectations about uncertain technological returns.

Let b_t represent collective beliefs about technological progress. Then

$$\pi_t^Q = \Phi(b_t, X_t, \mathcal{K}_t^Q) \quad (31)$$

represents the probability of investment or adoption.

Behavioural economics may therefore be relevant to the timing of investment, particularly where technological narratives generate optimism or excessive pessimism.

Psychiatry has no privileged role in diagnosing collective political decisions. Its relevant contribution here is methodological caution: individual psychological and psychiatric concepts should not be improperly generalized to nations, institutions or populations.

18 Philosophy and Ethics

Quantum infrastructure raises questions of intergenerational justice, security, privacy and responsibility.

A utilitarian formulation may consider

$$W = \sum_t \delta^t U_t. \quad (32)$$

A rights-based framework may instead impose constraints on permissible applications even where aggregate utility is positive.

The paper therefore treats technological capability and ethical permissibility as distinct variables.

$$\text{Technical Feasibility} \neq \text{Ethical Permissibility}. \quad (33)$$

19 Political Science and Governance

Large-scale quantum infrastructure requires institutions capable of allocating capital, coordinating research and maintaining public trust.

Let institutional quality be I_t . Then

$$\mathcal{K}_t^Q = K_Q(Q_t, H_t, E_t, C_t, S_t, F_t, I_t). \quad (34)$$

Governance affects procurement, research funding, intellectual property, security standards, education and international cooperation.

The governance problem is therefore not merely who owns a quantum computer, but who controls the institutional ecosystem required to make it productive.

20 Geopolitics and International Relations

Quantum infrastructure creates strategic interdependence.

For states a and b , define bilateral quantum dependence as

$$D_{ab}^Q = \omega_1 R_{ab} + \omega_2 H_{ab} + \omega_3 C_{ab} + \omega_4 F_{ab}, \quad (35)$$

where the components may represent research, human-capital, communications and financial dependencies.

International cooperation can increase capability through specialization:

$$\mathcal{K}_{\text{joint}}^Q > \mathcal{K}_a^Q + \mathcal{K}_b^Q \quad (36)$$

when complementarities are sufficiently large.

Conversely, excessive dependence on a single external supplier can reduce resilience.

The appropriate strategic objective is therefore not necessarily autarky, but resilient diversification.

21 Diplomacy

Quantum infrastructure creates opportunities for diplomatic cooperation in standards, research, education, cybersecurity and post-quantum migration.

A cooperative architecture may be represented as

$$G_{\text{international}}^Q = (V, E, W), \quad (37)$$

where countries are nodes and agreements, research collaborations, investment relationships and infrastructure connections are edges.

A country entering such a network contributes value when it supplies a capability that is insufficiently represented elsewhere.

This gives the SNoG a general principle:

$$\Delta \mathcal{K}^Q = \mathcal{K}_{\text{network}+\text{new}}^Q - \mathcal{K}_{\text{network}}^Q. \quad (38)$$

A proposed participant should demonstrate a positive and identifiable material contribution rather than merely a change of nationality.

22 Warfare Economics and Strategic Competition

Quantum infrastructure can affect strategic competition through cryptographic security, intelligence resilience, communications and economic productivity.

A stylized two-state differential game can be written as

$$\dot{Q}_{i,t} = \beta_i H_{i,t} + \chi_i I_{i,t} - \gamma_i Q_{i,t}, \quad (39)$$

with national objective

$$J_i = \int_0^T e^{-rt} [\Pi_i(Q_{i,t}, Q_{j,t}) - C_i(I_{i,t}, H_{i,t})] dt. \quad (40)$$

The purpose of this representation is to study strategic investment and technological competition. It does not prescribe military operations.

A central economic result is that human-capital investment can change the technological trajectory even when the physical hardware is unchanged.

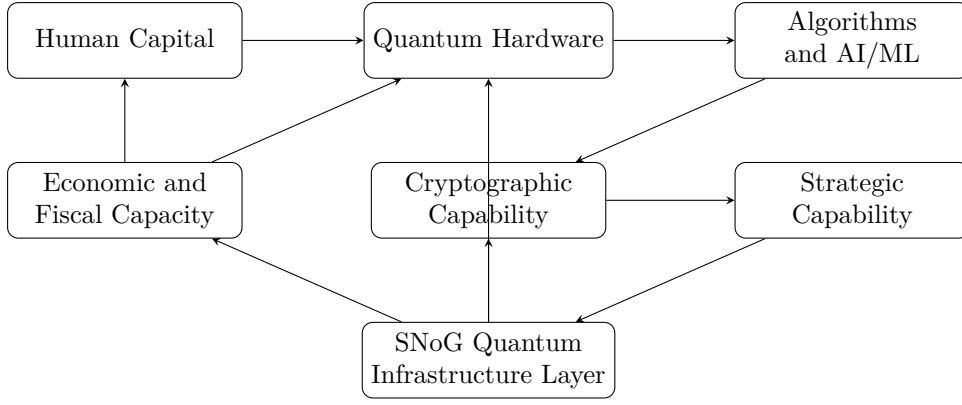


Figure 1: Architecture diagram.

23 The Quantum Infrastructure Feedback Loop

The complete SNoG mechanism can be represented as

$$\text{Human Capital} \rightarrow \text{Quantum Infrastructure} \rightarrow \text{Computational Capability} \rightarrow \text{Economic and Strategic Value} \rightarrow \text{Fiscal Capacity} \rightarrow \text{Further Investment} \quad (41)$$

The resulting feedback system is

$$\mathcal{Q}_{t+1} = F(\mathcal{Q}_t, H_t, F_t, X_t, G_t). \quad (42)$$

24 SNoG Architecture

The diagram emphasizes that the quantum infrastructure layer is not isolated. It is embedded within the broader economic and strategic infrastructure of the SNoG.

25 Algorithmic Implementation

A compact computational architecture is:

INPUT:

```

quantum hardware data
human-capital data
energy and communications data
financial data
algorithmic data
geopolitical data

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INITIALIZE:

```

quantum infrastructure network GQ
technological priors
real-option parameters

```

FOR each period t:

```

UPDATE human capital
UPDATE physical quantum capacity
UPDATE logical-error estimates
UPDATE algorithmic capability
UPDATE technological beliefs

```

```

ESTIMATE:
    quantum capability
    infrastructure option value
    resilience
    economic value
    strategic dependence

```

```

SIMULATE:
    technological shocks
    investment scenarios
    cooperation scenarios
    migration scenarios

```

```

UPDATE:
    fiscal capacity
    infrastructure portfolio
    international network

```

END FOR

```

RETURN:
    distribution of quantum capability
    infrastructure option values
    resilience measures
    policy scenarios

```

The architecture can be implemented using classical statistical models, Bayesian models, neural networks, graph neural networks or reinforcement learning, subject to validation and appropriate data.

26 Material Contribution Criterion

The central institutional principle of this paper is that a new country or institution should enter the SNoG quantum infrastructure network by demonstrating a material contribution.

Define

$$\Delta V_i = V(\mathcal{Q} \cup i) - V(\mathcal{Q}). \quad (43)$$

A contribution is materially differentiated when

$$\Delta V_i > 0 \quad (44)$$

and the increment arises from a capability not adequately represented by the existing network.

Possible sources include quantum hardware, error correction, algorithms, human capital, finance, energy, research institutions, industrial capacity, standards, cybersecurity, diplomatic connectivity or other infrastructure.

Nationality alone is therefore not a sufficient criterion.

27 The Italian Contribution

The Italy-specific contribution motivating this paper is the integration of quantum infrastructure with human-capital policy and cryptographic strategy.

The associated analysis identifies Italy's strategic quantum investments, neutral-atom capabilities and human-capital policies as mutually reinforcing components of a technological trajectory [1].

The proposed mechanism is

$$\text{Italian Human Capital} \rightarrow \text{Quantum Capacity} \rightarrow \text{Cryptographic Capability} \rightarrow \text{Strategic Infrastructure}. \quad (45)$$

This is materially different from simply adding another conventional infrastructure project to the SNoG.

28 Summary Table

Table 1: Summary table.

Layer	Infrastructure	Primary contribution
Physical	Quantum processors	Computational capacity
Error correction	qLDPC and related codes	Logical reliability
Human capital	Scientists and engineers	Skilled capacity
Software	Algorithms and AI/ML	Computational efficiency
Energy	Power and cooling systems	Operational capacity
Communications	Classical and quantum links	Network connectivity
Finance	Capital allocation	Investment capacity
Institutions	Universities and firms	Coordination capacity
Cryptography	Security infrastructure	Information security
Diplomacy	International networks	Strategic resilience
Welfare	Education and family policy	Human-capital formation

29 Conclusion

The Quantum Infrastructure of the Standard Nuclear oliGARCHy extends the general infrastructure theory by introducing a technological layer whose value depends simultaneously on physical systems, mathematics, algorithms, human capital, finance, institutions and international relations.

The central capability function is

$$\mathcal{K}_t^Q = K_Q(Q_t, H_t, E_t, C_t, S_t, F_t, I_t). \quad (46)$$

The central investment problem is a real-options problem:

$$V_t^Q = \max_{\tau} \mathbb{E}_t \left[e^{-r(\tau-t)} (P_{\tau}^Q - K_{\tau}^Q) \right]. \quad (47)$$

The central strategic principle is network complementarity:

$$\Delta V_i = V(\mathcal{Q} \cup i) - V(\mathcal{Q}). \quad (48)$$

A successful SNoG participant should therefore contribute a measurable capability that changes the system's feasible set, resilience or productive capacity.

The principal Italian contribution considered here is the connection between human-capital policy, quantum infrastructure and cryptographic capability. The resulting framework provides a bridge between economic development, technological competition, welfare economics, infrastructure theory and strategic resilience.

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Glossary

Quantum Infrastructure

The integrated physical, human, computational, financial, energy and institutional system required to produce effective quantum capability.

Physical Qubit

A physical implementation of a quantum two-level system used as a building block for quantum computation.

Logical Qubit

An error-corrected quantum information unit encoded across physical resources.

qLDPC

Quantum low-density parity-check error-correcting code.

Error Correction

The set of methods used to detect and suppress errors in quantum information.

Quantum Capability

The effective computational and technological capacity generated by the quantum infrastructure system.

Real Option

The value of having the ability to delay, expand, abandon or otherwise adapt an investment under uncertainty.

Quantum Human Capital

Scientific, engineering, mathematical and computational knowledge relevant to quantum technologies.

Cryptographic Infrastructure

The technological and institutional systems used to protect information and communications.

Network Complementarity

The increase in system value produced when connected infrastructure assets reinforce one another.

Post-Quantum Migration

The process of transitioning cryptographic systems toward security against relevant quantum attacks.

Strategic Resilience

The capacity of a state or network to maintain essential capabilities under technological, economic or geopolitical disruption.

Material Contribution

A measurable increase in the capability or resilience of an existing system caused by the addition of a new participant or infrastructure component.

SNoG

Standard Nuclear oliGARCHy, the broader systemic framework in which this quantum-infrastructure theory is situated.

The End